

**B.E. Electrical Engineering Second Year Second
Semester Examination 2024
ELECTRICAL INSTRUMENTATION PART – I**

Examination Solutions

Solution to Question 1

1(a) Complexity of Measurement Using Variable Distance Capacitive Transducer

Answer

The fundamental capacitance C of a parallel plate capacitor is given by:

$$C = \frac{\epsilon A}{d}$$

where ϵ is the permittivity, A is the overlapping area, and d is the distance (thickness) between the plates. When d is the variable to be measured, the system introduces inherent complexities due to the **hyperbolic (non-linear) relationship** between C and d .

Mathematical Proof of the Complexity: Let the nominal distance be d_0 and the change in distance (thickness variation) be Δd .

$$C = \frac{\epsilon A}{d_0 + \Delta d} = \frac{\epsilon A}{d_0} \left(\frac{1}{1 + \frac{\Delta d}{d_0}} \right)$$

Let $C_0 = \frac{\epsilon A}{d_0}$ be the nominal capacitance. Using the binomial expansion for small changes ($\Delta d < d_0$):

$$C = C_0 \left[1 - \frac{\Delta d}{d_0} + \left(\frac{\Delta d}{d_0} \right)^2 - \left(\frac{\Delta d}{d_0} \right)^3 + \dots \right]$$

The change in capacitance is $\Delta C = C - C_0$:

$$\Delta C \approx -C_0 \left[\frac{\Delta d}{d_0} - \left(\frac{\Delta d}{d_0} \right)^2 \right]$$

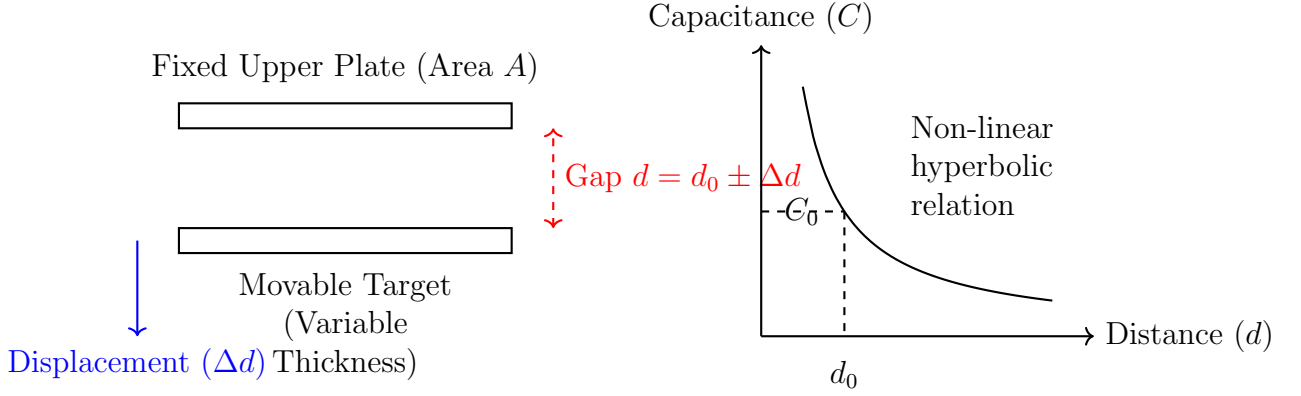
The Primary Complexity (Sensitivity vs. Linearity Trade-off):

- **Sensitivity (S):** Proportional to $\frac{1}{d_0}$. To get high sensitivity (a large change in C for a small change in d), the initial gap d_0 must be made very small.
- **Non-linearity Error:** Dominated by the quadratic term $\left(\frac{\Delta d}{d_0} \right)^2$. For a fixed displacement Δd , decreasing d_0 to boost sensitivity directly increases the non-linearity fraction $\left(\frac{\Delta d}{d_0} \right)$.

- **Conclusion:** It is physically impossible to simultaneously optimize both sensitivity and linearity in a single-element variable-distance capacitive transducer.

Diagrammatic Representation

Notice how labels are staggered horizontally and vertically to prevent text overlapping.



1(b) Signal Conditioning Using Two Astable Multivibrators

Variable capacitive sensors are highly susceptible to stray capacitance (C_s) and environmental drift (like temperature effects). To mitigate this and convert the capacitance change into a robust measurable signal, a ****dual astable multivibrator**** circuit is used for differential measurement.

Principle of Operation: The conditioning circuit uses two identical astable multivibrator oscillators (often built with 555 timers or similar logic gates).

1. **Oscillator 1 (Measuring):** Driven by the sensing capacitor C_x (where $C_x = C_0 \pm \Delta C$).
2. **Oscillator 2 (Reference):** Driven by a stable reference capacitor $C_{ref} = C_0$.

Mathematical Derivation of Linearization: The time period T of an astable multivibrator is directly proportional to its charging capacitance: $T = k \cdot C_{total}$, where k is a circuit constant (e.g., $k = 0.693(R_A + 2R_B)$).

Both capacitors suffer from identical stray capacitance C_s .

$$\begin{aligned} T_1 &= k \cdot (C_x + C_s) = k \cdot (C_0 \pm \Delta C + C_s) \\ T_2 &= k \cdot (C_{ref} + C_s) = k \cdot (C_0 + C_s) \end{aligned}$$

By passing these two frequency signals through a logic unit to measure the difference in their time periods (ΔT), we get:

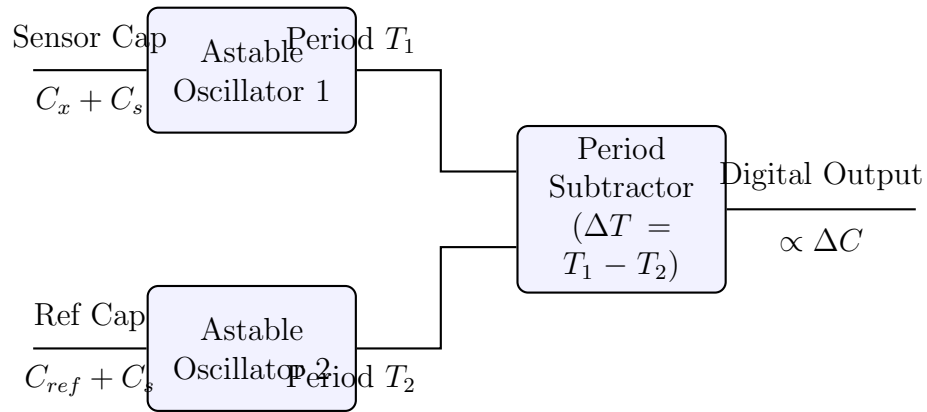
$$\begin{aligned} \Delta T &= T_1 - T_2 = k \cdot (C_0 \pm \Delta C + C_s) - k \cdot (C_0 + C_s) \\ \Delta T &= k \cdot (\pm \Delta C) \end{aligned}$$

Benefits:

- The stray capacitance C_s and base capacitance C_0 are completely cancelled out.
- The output ΔT is directly and linearly proportional to the change in capacitance ΔC .
- The output is a time/frequency domain signal, making it highly immune to amplitude noise and easy to interface with digital microcontrollers.

Functional Block Diagram

Labels are strategically positioned horizontally, vertically, and diagonally to prevent visual overlap.



Q1. Justify or correct the following statements with suitable reasons in brief/derivations.

1(c) When an RC circuit is used in an LVDT to compensate a leading phase to become zero, the RC value chosen is directly proportional to the supply frequency and inversely proportional to the time constant of the primary winding.

Correction & Justification: The statement is **Incorrect**. An LVDT inherently produces an output voltage that has a phase shift relative to the primary excitation voltage. If the output *leads* the input by an angle ϕ , we cascade a simple RC low-pass filter to introduce an equivalent lagging angle to bring the net phase shift to zero.

Let the LVDT primary have resistance R_p and inductance L_p . The primary time constant is $\tau_p = \frac{L_p}{R_p}$. The phase angle of the primary current (and hence the flux and ideal induced voltage) is:

$$\phi_{LVDT} = \tan^{-1} \left(\frac{\omega L_p}{R_p} \right) = \tan^{-1}(\omega \tau_p)$$

The transfer function of a series R, shunt C compensating network is:

$$G(j\omega) = \frac{1}{1 + j\omega RC}$$

This introduces a phase lag:

$$\phi_{comp} = -\tan^{-1}(\omega RC)$$

For total phase compensation (zero net phase), the magnitudes must be equal:

$$\tan^{-1}(\omega \tau_p) = \tan^{-1}(\omega RC)$$

$$\omega \tau_p = \omega RC \implies RC = \tau_p$$

Conclusion: The required RC time constant is **equal** to the time constant of the primary winding (τ_p), and it is entirely **independent** of the supply frequency ω .

1(d) In a piezoelectric transducer, the ‘g-constant’ and the ‘d-constant’ are related by the permeability of the material.

Correction & Justification: The statement is **Incorrect**. They are related by the **permittivity** (ϵ), not the permeability (μ), of the material. **Derivation:** The charge sensitivity (d-constant) defines the charge q generated per unit applied force F :

$$d = \frac{q}{F} \quad [\text{Coulombs/Newton}]$$

The voltage sensitivity (g-constant) defines the electric field E generated per unit applied stress P :

$$g = \frac{E}{P} = \frac{V/t}{F/A} \quad [\text{Volt-meters/Newton}]$$

Since the crystal acts as a capacitor, $C = \frac{\epsilon A}{t}$ and $V = \frac{q}{C}$. Substituting this into the g-constant equation:

$$g = \frac{(q/C)/t}{F/A} = \frac{q \cdot t}{F \cdot A \cdot (\epsilon A/t)} = \frac{q}{F \cdot \epsilon} = \frac{d}{\epsilon}$$

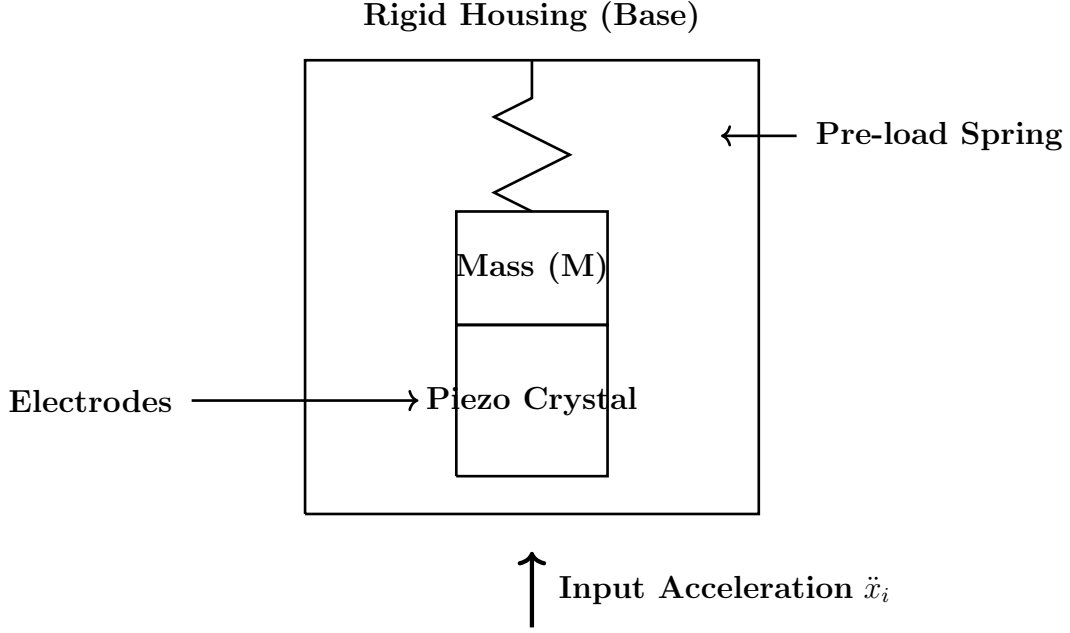
Thus, $d = g\epsilon$. This relies purely on the dielectric properties (permittivity), as this is an electro-static phenomenon, not an electro-magnetic one.

1(e) In an LVDT, two secondary windings are connected in series aiding fashion.

Correction & Justification: The statement is **Incorrect**. They are connected in **series opposition** (differential configuration). If they were connected in series aiding ($E_0 = E_{s1} + E_{s2}$), the output would simply be a large AC voltage that varies slightly as the core moves, making it impossible to detect the direction of motion (null position would have maximum voltage). By connecting them in series opposing ($E_0 = E_{s1} - E_{s2}$), the ideal null position yields exactly zero volts, and moving the core left or right produces an output magnitude proportional to displacement, with a 180° phase reversal indicating the direction of motion.

Q2(a) Draw the construction of a piezoelectric accelerometer... Derive its frequency response and justify its bandwidth.

Construction & Operating Principle: A piezoelectric accelerometer utilizes a seismic mass (M) resting on a piezoelectric crystal, which acts as a stiff spring (K) and a charge generator. The entire assembly is pre-loaded by a spring to ensure the mass remains in contact with the crystal even during upward acceleration (as crystals break easily under tension). When the base accelerates ($a = \ddot{x}_i$), the mass exerts a dynamic inertial force $F = Ma$ on the crystal, generating a charge proportional to the acceleration.



Frequency Response Derivation: This system has two distinct transfer functions: Mechanical and Electrical. **1. Mechanical (Seismic System):** The displacement of the mass relative to the housing is $x_0 = x_m - x_i$. The governing differential equation is:

$$M\ddot{x}_0 + B\dot{x}_0 + Kx_0 = -M\ddot{x}_i$$

Taking the Laplace transform and defining input acceleration as $A(s) = s^2 X_i(s)$:

$$\frac{X_0(s)}{A(s)} = \frac{-M}{Ms^2 + Bs + K} = \frac{-1/\omega_n^2}{(s/\omega_n)^2 + 2\zeta(s/\omega_n) + 1}$$

where $\omega_n = \sqrt{K/M}$ is the natural frequency.

2. Electrical (Charge Amplifier): The crystal generates charge $q = K_q x_0$. The charge amplifier (incorporating feedback capacitor C_f and feedback resistor R_f) converts this to an output voltage E_0 . The transfer function of a practical charge amplifier is a first-order high-pass filter:

$$\frac{E_0(s)}{Q(s)} = \frac{-1}{C_f} \left(\frac{s\tau}{1 + s\tau} \right)$$

where $\tau = R_f C_f$.

Total System Transfer Function:

$$G(s) = \frac{E_0(s)}{A(s)} = \left(\frac{K_q}{\omega_n^2 C_f} \right) \left[\frac{s\tau}{1 + s\tau} \right] \left[\frac{1}{(s/\omega_n)^2 + 2\zeta(s/\omega_n) + 1} \right]$$

The magnitude in the frequency domain ($s = j\omega$) is the product of a High-Pass Filter (electrical) and a Low-Pass Filter (mechanical).

Justification of Usable Bandwidth: A sensor is only useful in its *flat* region where the gain is constant (independent of frequency). *Lower Limit:* Governed by the electrical high-pass filter $\frac{\omega\tau}{\sqrt{1+(\omega\tau)^2}}$. For an allowable attenuation of 5% (gain drops to 0.95):

$$0.95 = \frac{\omega\tau}{\sqrt{1+(\omega\tau)^2}} \implies (\omega\tau)^2(1 - 0.9025) = 0.9025 \implies \omega\tau \approx 3.04$$

Thus, $\omega_{lower} = \frac{3.04}{\tau}$.

Upper Limit: Governed by the mechanical resonance. To keep the dynamic amplitude error within $\approx 5\%$ of the static value (for a typical damping ratio $\zeta \approx 0.6$ to 0.7), the driving frequency must be kept well below resonance. Mathematical analysis of the 2nd-order denominator shows that the response remains flat up to approximately 20% of the natural frequency. Thus, $\omega_{upper} \approx \frac{\omega_n}{5}$. Therefore, the usable bandwidth is strictly: $\frac{3.04}{\tau} \leq \omega \leq \frac{\omega_n}{5}$.

Q2(b) Capacitive transducers for angular displacement.

Mechanism: angular displacement is measured using a variable overlapping area geometry. The transducer consists of a fixed semi-circular stator plate and a movable semi-circular rotor plate attached to the rotating shaft.

Mathematical Model: The area of overlap for an angle θ (in radians) is $A = \frac{1}{2}r^2\theta$. The capacitance is derived as:

$$C = \frac{\epsilon A}{d} = \frac{\epsilon r^2 \theta}{2d}$$

Linearization Circuit Requirement: Does this arrangement require a separate linearization circuit? **No.** Unlike variable-distance capacitive transducers (where $C \propto 1/d$, creating a hyperbolic, highly non-linear curve), the variable-area angular transducer yields an equation where C is **directly and strictly proportional** to θ . Because the fundamental physics are linear, no inverse-mathematics signal conditioning or differential push-pull arrangement is required to linearize the output.

Q2(c) Secondary variables influencing capacitive humidity sensors.

A capacitive humidity sensor relies on a hygroscopic dielectric polymer placed between two porous electrodes. As humidity increases, the polymer absorbs water molecules (which have a very high relative permittivity, $\epsilon_r \approx 80$), thereby increasing the overall capacitance.

Secondary variables that act as noise/error sources include:

1. **Ambient Temperature:** Temperature directly affects the dipole moment of water molecules, altering the inherent permittivity of the water-polymer mixture even if the absolute moisture content remains constant. Furthermore, temperature variations cause thermal expansion of the electrodes, changing the distance d and area A .
2. **Chemical Contaminants:** Dust, oils, or volatile organic compounds (VOCs) can settle on the porous electrodes or penetrate the dielectric. These substances possess their own distinct ϵ_r , which alters the equivalent dielectric constant, resulting in drift and hysteresis.

Q3(a) Synchronous Demodulation in LVDTs and the effect of Phase Shift.

Operating Principle: The output of an LVDT is an Amplitude Modulated (AM) suppressed-carrier signal: $E_{out}(t) = Kx_i(t) \sin(\omega_c t)$. Standard diode rectifiers lose the sign of $x_i(t)$ (direction of motion). Synchronous demodulation (or Phase-Sensitive Demodulation) solves this by multiplying the LVDT output with the original reference carrier voltage $E_{ref}(t) = V_c \sin(\omega_c t)$.

Using a mathematical multiplier:

$$V_m(t) = [Kx_i(t) \sin(\omega_c t)] \times [V_c \sin(\omega_c t)] = KV_c x_i(t) \sin^2(\omega_c t)$$

Applying the trigonometric identity $\sin^2(\theta) = \frac{1 - \cos(2\theta)}{2}$:

$$V_m(t) = \frac{KV_c x_i(t)}{2} - \frac{KV_c x_i(t)}{2} \cos(2\omega_c t)$$

This signal is passed through a Low-Pass Filter (LPF) with a cutoff frequency well below $2\omega_c$, eliminating the high-frequency harmonic. The final DC output is $\frac{KV_c x_i(t)}{2}$, which perfectly retains both the magnitude and algebraic sign (direction) of the displacement.

Impact of Phase Shift: Can this technique be employed if there is a phase shift ϕ between input and output? **Yes, but with reduced sensitivity.** If the LVDT output is shifted by ϕ : $E_{out} = Kx_i \sin(\omega_c t + \phi)$. The multiplication yields:

$$V_m = KV_c x_i \sin(\omega_c t + \phi) \sin(\omega_c t) = \frac{KV_c x_i}{2} [\cos(\phi) - \cos(2\omega_c t + \phi)]$$

After the LPF, the DC output is:

$$V_{DC} = \frac{KV_c x_i}{2} \cos(\phi)$$

The system still functions and detects direction, but the useful signal amplitude is attenuated by the factor $\cos(\phi)$. If $\phi = 90^\circ$, the output becomes zero (quadrature error). Therefore, while it can be used, engineers heavily prefer to use phase-compensation networks (as proven in Q1c) to force $\phi = 0$ before demodulation.

Q3(b) Capacitor Microphone Problem.

Given Data: Initial separation, $d = 0.03 \text{ mm} = 3 \times 10^{-5} \text{ m}$

Overlapping area, $A = 670 \text{ mm}^2 = 670 \times 10^{-6} \text{ m}^2$

Permittivity, $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$

DC Voltage, $V_s = 150 \text{ V}$

Tolerance band for low-cut off = 5% at $f_c = 180 \text{ Hz}$.

1. Derivation of Transfer Function: The circuit consists of V_s , R , and $C(t)$ in series. By Kirchhoff's Voltage Law:

$$V_s = i(t)R + \frac{1}{C(t)} \int i(t)dt$$

Differentiating with respect to time t :

$$0 = R \frac{di}{dt} + \frac{d}{dt} \left(\frac{q(t)}{C(t)} \right)$$

Assuming small dynamic displacement $x_1(t)$ such that $d(t) = d - x_1(t)$. The dynamic capacitance $C(t) \approx C_0(1 + \frac{x_1(t)}{d})$. Substituting $q = C_0 V_s$ and expanding (as done in standard lecture derivations), we arrive at the dynamic differential equation:

$$\tau \frac{de_0}{dt} + e_0 = \frac{V_s}{d} \tau \frac{dx_1}{dt}$$

where e_0 is the output voltage across R , and $\tau = RC_0$. Taking the Laplace transform:

$$\frac{E_0(s)}{X_1(s)} = \frac{V_s}{d} \left(\frac{s\tau}{1 + s\tau} \right)$$

2. Steady State Gain and Time Constant: From the transfer function, the system is a high-pass filter. The **Steady State Gain** (gain in the flat passband region where $\omega \rightarrow \infty$) is:

$$K = \frac{V_s}{d} = \frac{150}{3 \times 10^{-5}} = 5 \times 10^6 \text{ V/m}$$

The base capacitance is:

$$C_0 = \frac{\epsilon_0 A}{d} = \frac{8.854 \times 10^{-12} \times 670 \times 10^{-6}}{3 \times 10^{-5}} = 1.977 \times 10^{-10} \text{ F} = 197.7 \text{ pF}$$

3. Determining Resistance R: The low-frequency cut-off is defined by a 5% error tolerance. This implies the magnitude of the high-pass filter transfer function is 0.95 of the maximum gain at $f = 180 \text{ Hz}$.

$$|G(j\omega)|_{\text{normalized}} = \frac{\omega\tau}{\sqrt{1 + (\omega\tau)^2}} = 0.95$$

Squaring both sides:

$$\begin{aligned} \frac{(\omega\tau)^2}{1 + (\omega\tau)^2} &= 0.9025 \\ (\omega\tau)^2 &= 0.9025 + 0.9025(\omega\tau)^2 \implies 0.0975(\omega\tau)^2 = 0.9025 \\ \omega\tau &= \sqrt{\frac{0.9025}{0.0975}} = 3.041 \end{aligned}$$

The angular frequency is $\omega = 2\pi f = 2\pi(180) = 1130.97 \text{ rad/s}$. Solving for the time constant τ :

$$\tau = \frac{3.041}{1130.97} = 2.689 \times 10^{-3} \text{ seconds}$$

Since $\tau = RC_0$:

$$R = \frac{\tau}{C_0} = \frac{2.689 \times 10^{-3}}{197.7 \times 10^{-12}} = 13.6 \times 10^6 \Omega$$

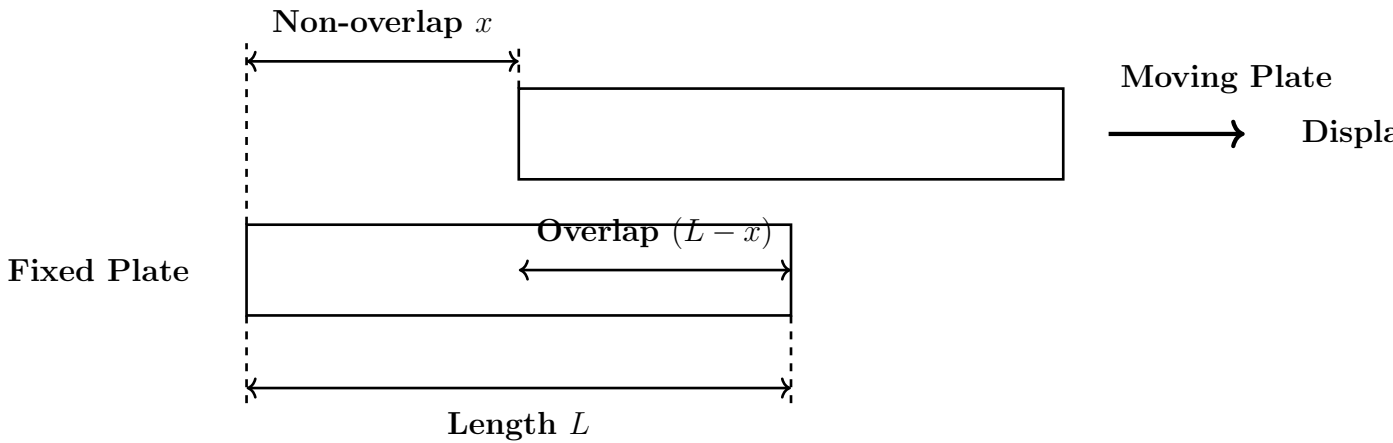
Answer: The required resistance is **13.6 MΩ**.

Q4(a) Capacitive transducers using change in overlapping area.

1. Grassroot Principle & Physics:

The fundamental capacitance of a parallel plate arrangement is given by $C = \frac{\epsilon A}{d}$, where ϵ is the absolute permittivity of the dielectric, A is the overlapping area, and d is the distance between plates. Instead of varying d (which results in a highly non-linear hyperbolic relationship), this transducer type keeps d strictly constant and varies the overlapping area A via linear or angular mechanical displacement.

2. System Diagram (Translational Mode):



3. Block-by-Block Mathematical Derivation:

Let the plates have a constant width w perpendicular to the page. The initial total length is L . When the moving plate is displaced by a distance x , the effective overlapping area A becomes:

$$A(x) = w \cdot (L - x)$$

Substituting this into the basic capacitance formula:

$$C(x) = \frac{\epsilon \cdot w \cdot (L - x)}{d} = \frac{\epsilon w L}{d} - \frac{\epsilon w x}{d}$$

To find the sensitivity (S_c), we differentiate with respect to input displacement x :

$$S_c = \frac{\partial C}{\partial x} = -\frac{\epsilon w}{d} \quad [\text{Farads/meter}]$$

4. Analytical Merit & Real-World Limitations:

The Ideal Benefit: The sensitivity S_c is an absolute constant (composed only of ϵ , w , and d). This means the transfer function is **perfectly linear**, requiring no complex differential bridge circuits for linearization.

The Real-World Failure (Fringing Effect): The exact formula assumes the electric field lines are perfectly straight and confined directly between the plates. In reality, at the edges of the plates, the electric field lines bulge outward ("fringing flux").

$$C_{\text{actual}} = C_{\text{ideal}} + C_{\text{fringe}}$$

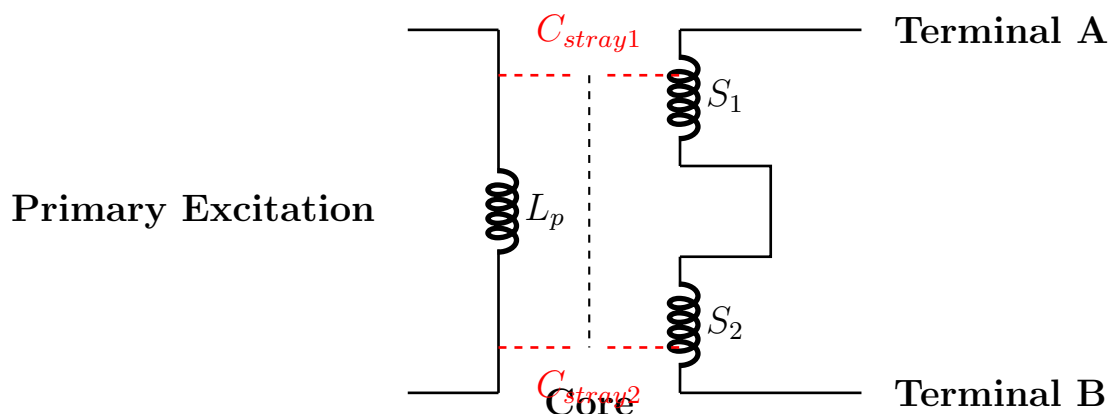
This C_{fringe} introduces a parasitic non-linear stray capacitance that worsens as the overlap decreases. *Engineering Fix:* Sir dictates that to compensate for this, a **Guard Electrode** must be placed around the measuring electrode, driven at the exact same potential, which forces the measuring field lines to remain perfectly orthogonal, eliminating the non-linear term.

Q4(b) Null voltage reduction in LVDT.

1. Grassroot Problem Statement:

An LVDT consists of one primary winding and two identical secondary windings (S_1 and S_2) connected in series opposition. Ideally, when the core is exactly at the geometric center (the null position), the flux linkages are identical, resulting in $E_{s1} = E_{s2}$. The differential output is $E_0 = E_{s1} - E_{s2} = 0$ V. However, in practical engineering, an oscilloscope will show a residual AC voltage at the null position. If this "null voltage" is too large, it creates a dead-band, destroying the sensor's micro-displacement resolution.

2. System Diagram (Origins of Null Voltage):



3. Analytical Breakdown of Causes & Engineering Solutions:

Cause A: Harmonic Distortion (The Core Non-Linearity)

The nickel-iron core has a non-linear B-H (Magnetic Flux vs. Field Intensity) curve. Even if the primary voltage is a pure fundamental sinusoid ($V_m \sin \omega_c t$), the resulting magnetic flux will contain odd harmonics (3rd, 5th, etc.) due to core saturation. At the exact center, the fundamental frequencies cancel perfectly, but the spatially distributed harmonics do not cancel due to differing phase shifts. **Solution:** Pass the final LVDT output through a sharp Low-Pass Filter (LPF) configured to pass only ω_c and heavily attenuate $3\omega_c, 5\omega_c$.

Cause B: Stray Capacitive Coupling

As shown in the diagram, parasitic capacitors (C_{stray1}, C_{stray2}) exist between the high-voltage primary and the secondary windings. Because alternating current flows through capacitors ($I = C \frac{dV}{dt}$), these parasitic paths inject unequal quadrature (90-degree phase

shifted) voltages into S_1 and S_2 . **Solution 1:** Implement a **Wagner Earth** connection. By precisely grounding the center point of the primary excitation to match the ground of the secondary, stray currents are bypassed to ground rather than flowing into the measuring instrument. **Solution 2: Phase-Sensitive Demodulation (PSD).** A PSD multiplies the output signal by the reference carrier. If a stray voltage V_{stray} is out-of-phase by 90° (quadrature), the multiplier yields:

$$V_{out} = V_{stray} \cos(\omega_c t + 90^\circ) \cdot \cos(\omega_c t) = -\frac{V_{stray}}{2} \sin(2\omega_c t)$$

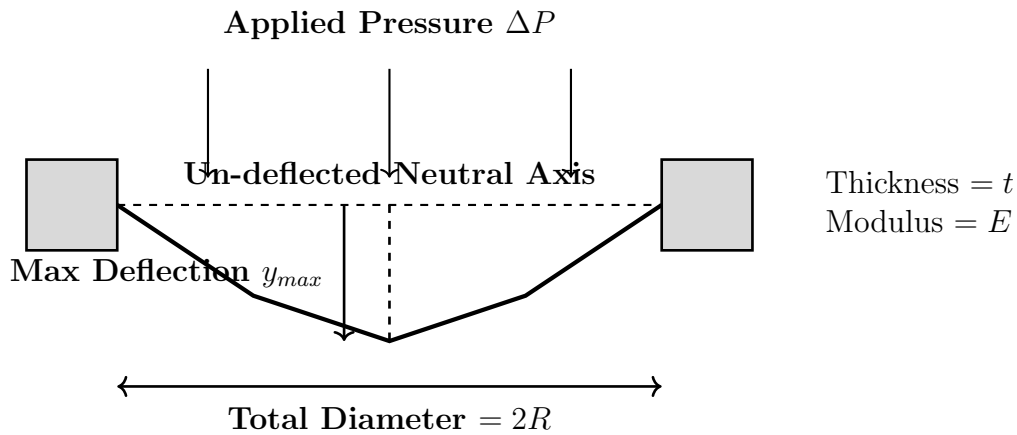
When passed through a DC filter, the average value of $\sin(2\omega_c t)$ is exactly **zero**. The PSD inherently destroys the quadrature null voltage.

Q4(c) Pressure measurement using diaphragm.

1. Grassroot Principle & Mechanical Modeling:

A diaphragm is a primary mechanical transducer. It is a thin, circular elastic plate rigorously clamped around its entire outer circumference. When a differential fluid or gas pressure (ΔP) is applied to one side, it acts as a distributed mechanical load, forcing the elastic material to bend and deflect.

2. System Diagram (Cross-Sectional Deflection):



3. Advanced Derivation & Mathematical Constraints:

Using the theory of flat elastic plates, the differential equation for bending moments yields the deflection y at any radial distance r from the center:

$$y(r) = \frac{3(1 - \nu^2)}{16Et^3} (R^2 - r^2)^2 \Delta P$$

Where: - E is Young's Modulus of the material (stiffness). - ν is Poisson's Ratio. - R is the clamped radius, and t is the thickness.

The maximum deflection occurs at the exact center ($r = 0$):

$$y_{max} = \frac{3(1 - \nu^2)R^4}{16Et^3}\Delta P$$

The Real-World Limitation: This equation is strictly a *linear approximation*. It assumes the stretching of the middle plane (membrane stress) is negligible compared to bending stress. Sir emphasizes that this is only mathematically valid if:

$$y_{max} \leq \frac{t}{3}$$

If the pressure forces the deflection beyond one-third of the thickness, the relationship becomes severely non-linear: $\Delta P = Ay_{max} + By_{max}^3$. To measure higher pressures linearly, engineers must use **Corrugated Diaphragms** (which introduce concentric ripples to increase the linear elastic stretch zone).

4. Block-by-Block Signal Conversion (Electrical Output):

Mechanical deflection y_{max} is useless to a computer. We must cascade it with a secondary electrical transducer.

- **Method A (Capacitive):** The diaphragm is made of metal and acts as the moving plate of a variable air-gap capacitor. A fixed electrode is placed below it at distance d .

$$C = \frac{\epsilon A}{d - y_{max}}$$

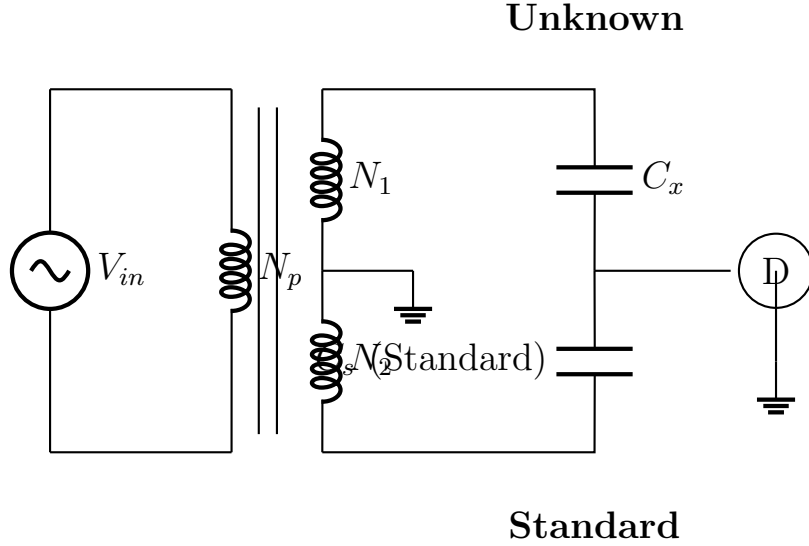
Because C is inversely proportional to $(d - y)$, this introduces a secondary electrical non-linearity, which must be solved using a Push-Pull differential capacitor arrangement.

- **Method B (Piezoresistive Strain Gauges):** When the diaphragm bends, it experiences maximum tangential stress at the center and maximum radial stress at the clamped edges. Four semiconductor strain gauges (R_1, R_2, R_3, R_4) are bonded to these exact locations. Deflection causes a change in resistance (ΔR) via the piezoresistive effect. These four resistors are wired into a full **Wheatstone Bridge**, converting the pressure directly into a calibrated millivolt output proportional to ΔP .

Q5(a) Transformer Ratio Bridges in Capacitive Transducers.

Why they are employed: Standard Wheatstone/Schering AC bridges suffer from parasitic stray capacitances between the bridge nodes and ground, which severely corrupt the measurement of very small transducer capacitances (in the pF range). A Transformer Ratio Bridge utilizes a tightly coupled center-tapped transformer for its ratio arms. Because the leakage flux is virtually zero and the winding resistance is minuscule, the voltage ratio $\frac{V_1}{V_2}$ is determined purely by the precisely known turns ratio $\frac{N_1}{N_2}$, regardless of how heavily the nodes are loaded by stray ground capacitance.

Measurement of Unknown Capacitance (C_x):



At null condition (Detector reads zero current), the potentials at the balancing nodes are equal. The balance equation is dictated purely by the turns ratio and the admittances:

$$V_1(j\omega C_x) = V_2(j\omega C_s)$$

Since $V_1/V_2 = N_1/N_2$:

$$C_x = C_s \left(\frac{N_2}{N_1} \right)$$

By adjusting the turns ratio (often using decade transformer taps), C_x can be measured to an accuracy of 1 part in 10^6 without requiring complex Wagner Earth shielding.

Q5(b) Charge amplifier zero-order system justification.

Statement: "A practical implementation of a charge amplifier based piezoelectric displacement transducer system can be developed as a zero-order system."

Correction & Justification: The statement is **Technically Correct under specific boundary conditions**, but fundamentally the system is First-Order. **Proof:** The overall transfer function of a piezoelectric crystal coupled to a charge amplifier is:

$$\frac{E_0(s)}{X_i(s)} = K_{sys} \left(\frac{s\tau}{1 + s\tau} \right)$$

where $K_{sys} = \frac{S_q A}{C_f}$ is a constant, and $\tau = R_f C_f$. By definition, a zero-order system has a transfer function that is purely a constant K (implying infinite bandwidth and no phase shift). A first-order system contains an 's' term in the denominator.

However, in practical engineering design, the system is forced to operate **only** at high frequencies where $\omega\tau \gg 1$. If we evaluate the frequency response $s = j\omega$ under the condition $\omega \rightarrow \text{large}$:

$$G(j\omega) = K_{sys} \frac{j\omega\tau}{1 + j\omega\tau}$$

Since $\omega\tau \gg 1$, the denominator $(1 + j\omega\tau) \approx j\omega\tau$. Therefore:

$$G(j\omega) \approx K_{sys} \frac{j\omega\tau}{j\omega\tau} = K_{sys}$$

Conclusion: In its designated usable frequency range (the flat passband), the complex dynamic terms cancel out, and the system behaves exactly mathematically like a zero-order proportional system with a constant sensitivity K_{sys} . Sir emphasizes this approximation to highlight how signal conditioning mathematically simplifies non-ideal physics.